across all stocks. However, proper statistical analysis is still required to ensure accurate results.

Factor models provide better insight into the overall covariance and correlation structure between stocks and across the market. Positive correlation means that stocks will move in the same direction and negative correlation means that stocks will move in opposite direction.

A factor model has the form:

$$r_t = \alpha_0 + f_{1t}b_1 + f_{2t}b_2 + \cdots + f_{kt}b_k + e_t \tag{6.32}$$

where,

r_t = stock return in period t.
α_0 = constant term.
f_{kt} = factor k value in period t.
b_k = exposure of stock i to factor k. This is also referred to as beta, sensitivity, or factor loadings.
e_t = noise for stock i in period t. This is the return not explained by the model.

Parameters of the model are determined via ordinary least squares (OLS) regression analysis. Some analysts apply a weighting scheme so more recent observations have a higher weight in the regression analysis. These weighting schemes are often assigned using a smoothing function and "half-life" parameter. Various different weighting schemes for regression analysis can be found in Green (2000).

To perform a statistically correct regression analysis the regression model is required to have the following properties. (see Green (2000); Kennedy (1998); Mittelhammer (2000), etc.).

Regression properties:

1. $E(e_t) = 0$
2. $Var(e_t) = E(e'e) = \sigma_e^2$
3. $Var(f_k) = E[(f_k - \bar{f}_k)^2] = \sigma_{fk}^2$
4. $E(ef_k) = 0$
5. $E(f_{kt}, f_{lt}) = 0$
6. $E(e_t e_{t-j}) = 0$
7. $E(e_{it} e_{jt}) = 0$

Property (1) states that the error term has a mean of zero. This will always be true for a regression model that includes a constant term $\hat{b}_{0k}$ or for a model using excess returns $E(r_{it}) = 0$. Property (2) states that the variance